

SET 1

1. R: Create 10 student marks and calculate mean, median, variance and standard deviation.

The screenshot shows the R Studio interface. The Console pane on the left displays the following R code and its output:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

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Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> marks <- c(78, 85, 67, 92, 74, 88, 81, 69, 95, 76)
>
> mean(marks)
[1] 80.5
> median(marks)
[1] 79.5
> var(marks)
[1] 89.16667
> sd(marks)
[1] 9.44281
>
> |
```

The Environment pane on the right shows the 'marks' variable as a numeric vector of length 10, with values: 78, 85, 67, 92, 74, 88, 81, 69, 95, 76.

The Files pane at the bottom shows a list of files in the 'Home' directory, including 'Rhistory', 'Analytical problems - CO2.docx', 'APPLICATION BASED QUESTIONS - M. Vineela.docx', 'APPLICATION BASED.docx', 'BEEE CO2 AT1 (1).docx', and 'BEEE- CO3- AT1.docx'.

2. R: For 12,18,21,25,29,31,35,40,44,50 find Q1,Q2,Q3,IQR and draw a boxplot.

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```
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Type 'q()' to quit R.

> x <- c(12,18,21,25,29,31,35,40,44,50)
>
> Q1 <- quantile(x, 0.25)
> Q2 <- quantile(x, 0.50)
> Q3 <- quantile(x, 0.75)
> IQR <- Q3 - Q1
>
> Q1
25%
22
> Q2
50%
30
> Q3
75%
38.75
> IQR
75%
16.75
>
> boxplot(x,
+         main="Boxplot of Student Data",
+         ylab="Values")
> |
```

The Environment pane on the right shows the 'x' variable as a numeric vector of length 10, with values: 12, 18, 21, 25, 29, 31, 35, 40, 44, 50. It also shows the calculated quartiles and IQR:

Variable	Value
IQR	Named num 16.8
Q1	Named num 22
Q2	Named num 30
Q3	Named num 38.8
x	num [1:10] 12 18 21 25 29 31 35 40 44 50

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The boxplot titled "Boxplot of Student Data" is displayed in the bottom right pane. The y-axis is labeled "Values" and ranges from 20 to 50. The boxplot shows the distribution of the data, with the median (Q2) at 30, the first quartile (Q1) at 22, and the third quartile (Q3) at 38.75. The whiskers extend from the box to the minimum and maximum values of the data.

3. WEKA: Create an ARFF customer dataset (Age, Income, Purchase) and apply J48; visualize the tree and record correct/incorrect instances.

Dataset: Customer Purchase

```
@relation customer_purchase
@attribute Age numeric
@attribute Income numeric
@attribute Purchase {Yes,No}
@data
22,25000,No
25,32000,No
28,45000,Yes
31,52000,Yes
35,60000,Yes
40,72000,Yes
23,28000,No
29,48000,Yes
```

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: customer_purchase Attributes: 3 Instances: 8 Sum of weights: 8

Selected attribute: Name: Income Missing: 0 (0%) Distinct: 8 Type: Numeric Unique: 8 (100%)

Statistic	Value
Minimum	25000
Maximum	72000
Mean	45250
StdDev	16342.32

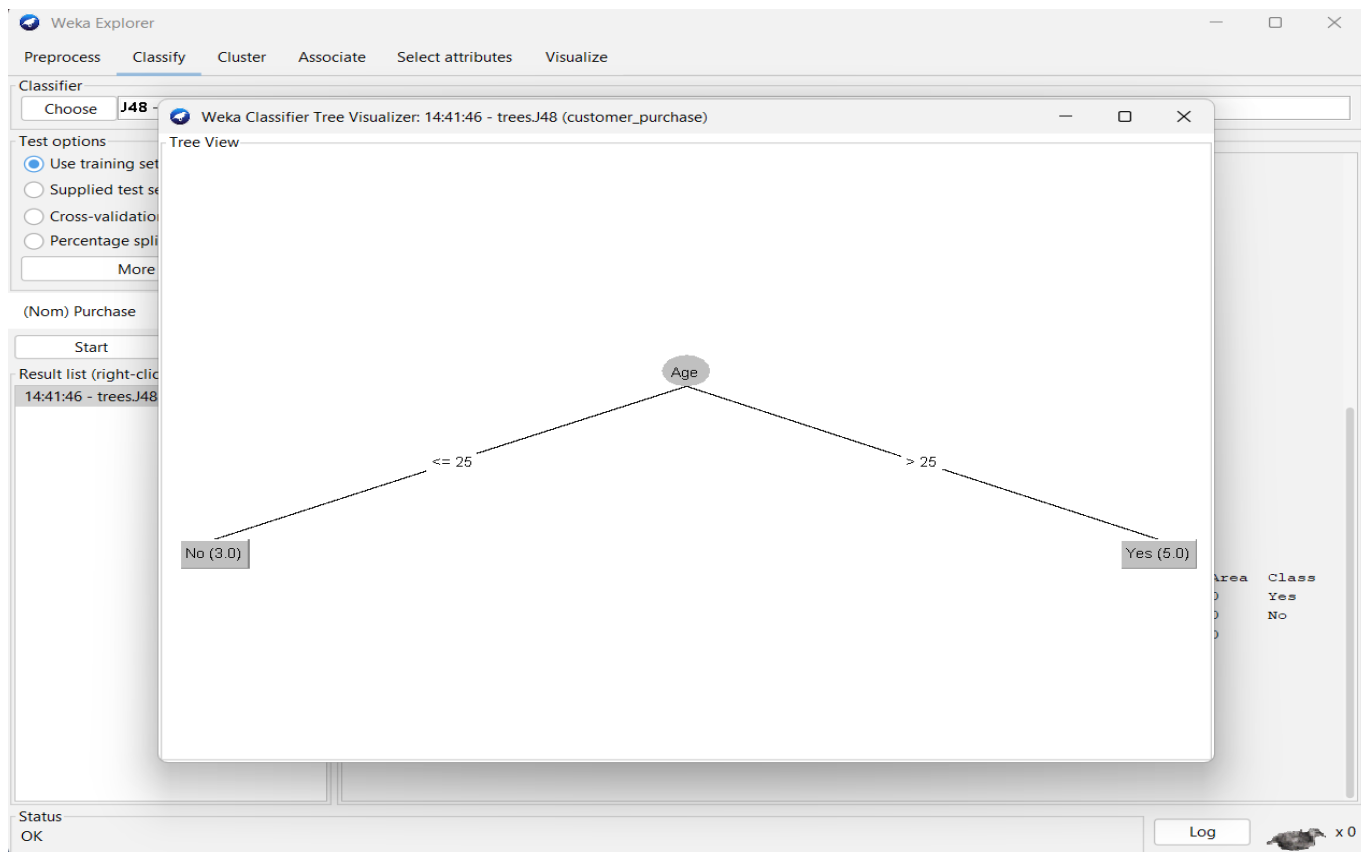
Class: Purchase (Nom) Visualize All

6 3 25000 48000 72000

Status: OK Log x 0

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4. WEKA: Apply Naive Bayes to a suitable nominal dataset and report accuracy, precision, recall and F1-score.

Use the same Customer Purchase ARFF dataset given for Set 1 Q3.

